

RT-IOT2022



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RT-IOT2022 Dataset

Descriptor	Value
Feature Count	83
Size	53MBytes
Row Count	123,118
Missing Values	No

Problem

Can a Decision Tree classify traffic based on the training data in the dataset?

Makefile

- Supports rapid iteration
- Checks for proper configuration
- Stages and retrieves data from HDFS
- Proper dependency checking to avoid re-execution

```
$ make all
```

```
$ make clean
```

Preprocessing

- Remove row_id field
- Rename fields with “_” in them
- StringIndex text fields and the label

TargetEncoding

- Encodes a {column, target} pair into a likelihood
- Allows categorical fields (text, port numbers) to be standardized and passed to PCA
- Converts a categorical field to the range $[0, 1]$

Cluster vs Local

- Single baseline pass on a laptop took 15s
- Single baseline pass in the cluster took 75s
- Cluster was unable to run 10 model trainings in parallel

Results

Label	Precision	Recall
NMAP TCP scan	0.30	0.003
NMAP OS DETECTION	0.43	0.69
NMAP FIN SCAN	0.00	0.00
Metasploit Brute Force SSH	0.00	0.00
MQTT Publish	1.00	1.00

Conclusion

- Spark is a tool used only when it is “necessary”
- DecisionTree doesn't work well with this dataset
- A pipeline that tests all model types would be useful